

Elvis F. Elli

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Sent: Friday, July 31, 2026 3:51 PM
To: Elvis F. Elli
Subject: Review Package for Proposal 2025-13716 Submitted to USDA\NIFA

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July 31, 2026

Area: Agriculture and Food Research Initiative

Program: Plant Health and Production and Plant Products: Plant Breeding for Agricultural Production

Proposal Number: 2025-13716

Project Director: Elvis Elli

Proposal Title: Integrating genomic prediction and crop growth models to maximize soybean seed yield potential

Dr. Elvis Elli

Division of Agriculture of the University of Arkansas System

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Dear Dr. Elli,

Your proposal submitted to the USDA\NIFA AFRI A1141 Plant Breeding for Agricultural Production program has completed peer review. Attached is your review package, which includes the panel's comments and the relative ranking of your application.

We regret to inform you that we are unable to provide funding support for your project this year. The A1141 program area priority received an exceptionally high volume of applications, and not all proposals that received competitive rankings could be supported.

If you plan to resubmit, we encourage you to review the updated NOFO when it is released and carefully consider the panel's feedback as you prepare future submissions.

Thank you for your interest in this program.

Regards,

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United States Department of Agriculture
National Institute of Food and Agriculture
Agriculture and Food Research Initiative
Plant Health and Production and Plant Products: Plant Breeding for Agricultural Production

Proposal 2025-13716 submitted by Elli

Integrating genomic prediction and crop growth models to maximize soybean seed yield potential

The review panel grouped proposals into one of the relative categories below. The percentage indicates the final distribution of proposals in each category.

Outstanding %	20
High Priority %	28
Medium Priority %	12
Low Priority %	40
Do Not Fund %	0

This proposal was placed in : High Priority

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Plant Health and Production and Plant Products: Plant Breeding for Agricultural Production - PANEL SUMMARY

The panel decision regarding your proposal is based on the input provided by the reviews and the collected expertise and judgment of the individual panel members. This panel summary reflects the consensus opinion of the panel regarding your proposal.

Proposal Number: 2025-13716 Project Director: Elli

Proposal Title: Integrating genomic prediction and crop growth models to maximize soybean seed yield potential

Positive Aspects of the Proposal

Crop growth models (CGM) are of a lot of interest in literature right now as scientists seek the next step change beyond GEBVs to unlock genetic gain. Nice use of preliminary results from APSIM runs. Benchmarking against other genomic prediction models will show potential relevance to this approach. The University of Arkansas runs a breeding program that gives them real germplasm to use and an obviously relevant user of the results. Field plan includes a mix of frequent human-phenotyping and UAVs to closely monitor growth. The team is well-situated to conduct the proposed work.

Negative Aspects of the Proposal

The proposed study may be limited by its scope, as trials at only two locations (Arkansas and Missouri) with two replications may not adequately capture genotype × environment (G×E) interactions, making it difficult to draw robust conclusions. Extrapolating phenotypic data to other environments may be challenging due to the limited range of conditions represented in the training set. It is also unclear whether these environments will provide sufficient variation to distinguish genetic effects from environmental and G×E influences. While focusing on maturity zones 3 and 4 may be reasonable given budget constraints, it could limit the generalizability of the model and risk overfitting.

It is interesting that epigenetics is mentioned in the rationale, however, activities involving epigenetics are not described in the proposal. The genotyping platform that is planned for use in genomic prediction also does not appear to have been mentioned.

It seems worth considering a potential pitfall if the crop growth model predicts yield fairly accurately without accurately predicting phenology (or vice versa).

Predicting 3,000 genetically unrelated lines seems risky as the marker effects may not be consistent across the training and prediction sets, but this part of the analysis (Task 3.1: Establish a benchmark analysis of the genomic prediction-based crop modeling framework) was described only briefly.

Citations are missing in a couple of key places.

Synthesis Comments

This proposal is well-written and relevant to the program. Overall, describing the genotyping platform, how a suited training set would be established for the 3,000 individuals that were mentioned, and considering expanding the maturity zones in which empirical evaluation (field trialing and model evaluation) is conducted, especially if wanting to examine interactions between photoperiod and temperature, would be helpful.

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The following reviews were submitted for your proposal, the names of the reviewers have been removed to maintain confidentiality.

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This proposal was well-written (though missing citations in a couple of key places where noted below). The team seems well-situated to conduct the proposed work. The focus on maturity zones 3 and 4 seems a bit narrow (and could also limit generalizability of the model) but is also within reason given the budgetary scope.

This sentence is missing a few key references from Corteva (starting in 2015) that integrated crop growth modeling and genomic prediction: “Studies integrating genomic-based predictions and crop modeling have been reported for multiple crops with varying levels of success (Bogard et al., 2014; Laurent et al., 2025; Messina et al., 2006; Oliveira et al., 2021; Vallejos et al., 2022; White and Hoogenboom, 1996).” There was an announcement during the review period that the source code for that framework from Corteva is now available for academic, non-commercial use, in case of interest.

Using an ‘unexplored genetic pool’ sounds difficult, as fairly substantial data can be needed to derive genotype-specific coefficients in crop growth models, and to ensure they are truly genetic parameters rather than being influenced by E and/or GxE.

Nice use of preliminary results from APSIM runs.

Interesting that epigenetics is mentioned as being in scope, as activities involving epigenetics are not described in the proposal. The genotyping platform that is planned for use in genomic prediction also does not appear to have been mentioned.

Regarding this sentence, “Previous research has shown important but limited information on photothermal interactions modulating the possible extent of exploitable genetic variation in soybean phenology (Messina et al., 2006).”: It seems some additional relevant work on photothermal interactions in soybean would have been conducted within the past 20 years? For example: McCormick et al. 2021, ‘Intercontinental prediction of soybean phenology via hybrid ensemble of knowledge-based and data-driven models’, <https://doi.org/10.1093/insilicoplants/diab004>.

A related note: a quick search of “soybean photothermal response” in Google Scholar gave rise to at least five relevant papers within the first two pages of search results, none of which were cited in this proposal. Those papers are listed below (using the reference format of each publisher’s site). It seems a more comprehensive literature review would be important.

Wu, T., Lu, S., Cai, Y. et al. Molecular breeding for improvement of photothermal adaptability in soybean. *Mol Breeding* 43, 60 (2023). <https://doi.org/10.1007/s11032-023-01406-z>

Xintian Zhu, Willmar L Leiser, Volker Hahn, Tobias Würschum, The genetic architecture of soybean photothermal adaptation to high latitudes, *Journal of Experimental Botany*, Volume 74, Issue 10, 19 May 2023, Pages 2987–3002, <https://doi.org/10.1093/jxb/erad064>

Wu, T., Wen, H., Zhang, X. et al. Genome-wide association study for temperature response and photothermal interaction of flowering time in soybean using a panel of cultivars with diverse maturity groups. *Theor Appl Genet* 136, 245 (2023). <https://doi.org/10.1007/s00122-023-04496-7>

Schoving C, Stöckle CO, Colombet C, Champolivier L, Debaeke P and Maury P (2020) Combining Simple Phenotyping and Photothermal Algorithm for the Prediction of Soybean Phenology: Application to a Range of Common Cultivars Grown in Europe. *Front. Plant Sci.* 10:1755. <https://doi.org/10.3389/fpls.2019.01755>

Mao, T., Li, J., Wen, Z. et al. Association mapping of loci controlling genetic and environmental interaction of soybean flowering time under various photo-thermal conditions. *BMC Genomics* 18, 415 (2017). <https://doi.org/10.1186/s12864-017-3778-3>

It is not clear that the planned field environments in Arkansas and Missouri will expose relevant variation for the disentanglement of genetic parameters vs. E- and/or GxE-influenced parameters. However, if a main goal is for the developed model to work well within maturity zones 3 and 4, this trialing configuration could be sufficiently suited. Plans to broaden out to include other maturity zones would be helpful as a follow-on (and it seems good to avoid overfitting to zones 3 and 4, and/or overestimating the heritability of genetic parameters, in the meantime).

Regarding Task 2.1: The flexibility of using covariates is mentioned. Do you plan to use covariates while estimating the genotype-specific coefficients? If so, which covariates are planned for use? It had seemed the goal would be to focus on estimating genetic parameters (or ‘genotype-specific coefficients’) that are thought to not be influenced by environmental covariates (for instance). While a genetic parameter itself would not be influenced prior to input into the model, it could instead be modified by some E- or M-influenced parameter (or ‘covariate’) after having been inputted into the crop model. However, if the project team is proposing a hybrid between the use of environmental covariates in genomic prediction and the use of integrated crop growth modeling and genomic prediction, further elaboration on this point would be interesting.

It seems perhaps worth considering an approach in which days to flowering, days to maturity, and/or seed yield are inputted to the model (whether observed or predicted via genomic prediction), and genotype-specific parameters are then estimated if only allowing a subset of those parameters to vary. Just in case this complementary approach is in any way informative. It also seems worth considering a potential pitfall of the crop model predicting yield fairly accurately without predicting days to flowering accurately, or vice versa, and how important the accurate prediction of both of those traits alongside days to maturity is to your overall approach.

Including more than two replicates in field trialing seems helpful if feasible (even if only collecting detailed phenology-relevant data in two replicates per planting date), given the interest in yield. Nice inclusion of two planting dates.

Predicting 3,000 genetically unrelated lines seems risky as the marker effects may not be consistent across the training and prediction sets, but this part of the analysis (Task 3.1) was described only briefly.

Scientific Merit

The overall goal of this project is to identify strategies to enhance soybean seed yield potential by integrating genomic prediction with crop growth modeling. The central hypothesis is that a consistent yield advantage can be achieved by optimizing phenological life-cycle traits across the growing season to maximize light interception and extend the seed-filling period. The specific objectives are to: (1) quantify the effects of temperature and photoperiod on developmental rates among diverse soybean accessions; (2) develop a genomic prediction-based crop modeling framework by incorporating genomic prediction models into the APSIM-Soybean crop growth model; and (3) identify soybean ideotypes for target environments using this integrated modeling framework.

The proposal is supported by a comprehensive methodology designed to achieve the three specific objectives. Each objective is well-structured and includes background, approach, expected results, potential pitfalls, and alternative strategies. The proposal also includes sections detailing dissemination plans and a project timeline. The expected outcomes include: (1) quantitative data on the contributions of temperature and photoperiod to variability in soybean phenological traits; (2) the development of a genomic prediction-based crop modeling framework to enhance the linkage between soybean physiological traits and their genetic regulation; and (3) the identification of soybean ideotypes for target environments using this integrated framework.

Qualifications of Project Personnel, Adequacy of Facilities, and Project Management

The applicants have demonstrated the project's feasibility through preliminary research and access to well-equipped facilities. The data management plan describes expected data types and formats, as well as strategies for data storage, preservation, sharing, protection, and public access. It further outlines the roles and responsibilities of the investigators in data management, along with the relevant experience and duties of key personnel in the project. A comprehensive mentoring plan is also included for graduate, undergraduate, and postdoctoral researchers, addressing their needs in research skill development, professional development, scientific communication, and career preparation.

Project Relevance

This work advances AFRI Priority A1141, leveraging plant genetic resources by investigating diverse soybean accessions with known genetic information to better understand the genetic control of phenology and knowledge into predictive modeling frameworks to improve selection for high-yielding genotypes. It supports breeding efforts to improve crop productivity, efficiency, and performance.

This project aims to use crop growth modeling to better understand/characterize soybean yield potential and in turn describe ideotypes that can be leveraged in breeding to improve genetic gain. They hypothesize that they can extend this model into predictions specific to environments which would be powerful.

Relevance

Strengths

Letters of support from state department of agriculture. CGM is also of a lot of interest in the literature right now as scientists seek the next step change beyond GEBVs to unlock genetic gain.

Evidence of variability for phenology among relevant germplasm and likely relevant traits

University of Arkansas runs a breeding program which gives them real germplasm to use and a obviously relevant user of the results.

Benchmarking against other genomic prediction models will show potential relevance of this approach

Weaknesses

Does APSIM allow you to look at feature importance to determine what environment or physiological factor is most important?

Feasibility

Strengths

Field plan includes a mix of frequent human-phenotyping and UAVs to closely monitor growth. 2 locations (with 2 reps) is small given the amount of GxE with these traits and may not be sufficient to draw conclusions.

APSIM-Soybean is a validated model used by other programs.

Weaknesses

It may be difficult to extrapolate phenotypes from the limited phenotyping environments to additional locations even with strong weather/soil data as those environments may not be represented in your training set.

There are two PIs leading phenotyping at different farms. This is a strength because time intensive phenotyping will be more practical when the team is close by, but care must be taken to have observations concordant across the two teams.

Qualification of PI & Budget Comments

Strengths: qualified PIs who are collocated with the phenotyping locations.